Farshid Farshidfar

Mohamadreza Farshidfar in official records

Mathematics PhD researcher | University of Sussex

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Profile

Mathematics PhD researcher working across probability, geometry, topology and mathematical physics. I use lattice models, random matrices and representation theory to study two-dimensional Yang–Mills theory and large-N limits. My background includes symplectic geometry, Floer theory and dynamical systems; my wider interests include the mathematical foundations of machine learning.

Professional experience

Mar 2026–present

Scientific Reviewer for AI-Generated Mathematics and Physics

PI1 Labs | Remote

Review research-level mathematics and theoretical-physics content generated by AI systems, assessing correctness, assumptions, logical consistency and rigour. Identify errors and gaps in technical arguments and independently evaluate whether conclusions are mathematically justified across advanced topics.

Education

2023–present

PhD in Mathematics

University of Sussex, Brighton, UK

Supervisor: Antoine Dahlqvist.

2021–2022

Master of Science in Mathematics (M2)

Joint programme: Sorbonne Université, Université de Paris and Université Sorbonne Paris Nord, France

2012–2017

BSc in Mathematics and Applications

Sharif University of Technology, Tehran, Iran

Current research and manuscripts

Manuscripts in final preparation; arXiv posting anticipated by December 2026 (estimate as of October 2026).

Yang–Mills measure and the master field on the real projective plane.

With Antoine Dahlqvist.

Large-N asymptotics of Wilson loop observables for $U(N)$ Yang–Mills theory on the real projective plane. Methods include heat-kernel lattice measures, character expansions, Frobenius–Schur indicators and Makeenko–Migdal equations, with separate analysis of contractible and noncontractible loops.

Atiyah–Bott–Goldman measure and the low-area limit of two-dimensional quantum Yang–Mills measure.

With Antoine Dahlqvist.

Small-area limits of lattice Yang–Mills measure for compact connected semisimple groups on closed oriented surfaces of genus $g \geq 2$. Study of the relation to normalized Atiyah–Bott–Goldman Liouville measure on the smooth top stratum of the character variety, using generalized Makeenko–Migdal equations, Goldman's Poisson bracket and Hamiltonian invariance.

Yang–Mills theory on higher-genus non-orientable surfaces.

With Antoine Dahlqvist.

Study of large-N Wilson loop limits on higher-genus non-orientable surfaces, focusing on $U(N)$ and odd non-orientable genus at least three. Methods include representation-theoretic character estimates, loop topology, Makeenko–Migdal equations and comparison with planar master-field candidates on the universal cover.

Earlier research experience

2022 — **Feral curves and minimal sets**
Sorbonne Université | Supervisor: Vincent Humilière
Master's thesis reviewing the paper by Joel W. Fish and Helmut Hofer; prepared a memoir explaining its proof.

2018–2019 — **Floer homology and time-delayed dynamics**
Université de Neuchâtel, Switzerland | Supervisor: Felix Schlenk
Studied compactness, bubbling, Fredholm theory and gluing for pseudoholomorphic curves, and introductory polyfold theory. Prepared a memoir on Fredholm properties of Floer equations with a time-delayed action functional.

2014 — **Equivariant cohomology and the Lefschetz fixed-point theorem**
Sharif University of Technology | Supervisor: Mostafa Esfahani Zadeh
Studied analysis on manifolds, the Laplacian on differential forms and heat flow, with an exploratory project on fixed orbits of Lie group actions.

Talks

2024 — **Two-dimensional quantum Yang–Mills theory**
University of Reading, UK

2024 — **Large deviations principle for the Curie–Weiss model**
University of Sussex, UK

2022 — **Feral curves and minimal sets**
Sorbonne Université, France

2022 — **Combinatorial applications of the Borsuk–Ulam theorem**
Université Sorbonne Paris Nord, France

2019 — **Supersymmetry, quantum tunnelling and Morse theory**
Institute for Research in Fundamental Sciences (IPM), Tehran, Iran

2019 — **Fredholm property of the Floer equation**
Université de Neuchâtel, Switzerland

Spring 2018 — **What if Alice and Bob had a quantum computer?**
A protocol for transmitting quantum data; Université de Neuchâtel, Switzerland

Spring 2018 — **Periodicity of delayed population dynamics**
Université de Neuchâtel, Switzerland

Autumn 2017 — **A Hamiltonian with no periodic orbits in an infinite-dimensional Hilbert space**
Université de Neuchâtel, Switzerland

Summer 2016 — **Symplectic packing**
Sharif University of Technology, Iran

Spring 2015 — **Geodesic flow and horocycle flow**
Sharif University of Technology, Iran

Spring 2014 — **The Lefschetz fixed-point theorem via heat flow**
Sharif University of Technology, Iran

Autumn 2013 — **Constant mean curvature surfaces with boundary and the tangency principle**
Sharif University of Technology, Iran

Autumn 2013 — **The exponential map and properties of geodesics on surfaces**
Sharif University of Technology, Iran

Teaching

Autumn 2014 — **Teaching assistant: Differential Geometry of Curves and Surfaces**
Sharif University of Technology
Instructor: Dr Bahreini.

Spring 2014 — **Teaching assistant: Analysis II**
Sharif University of Technology
Instructor: Mohsen Sharifitabar.

2013 — **Differential geometry for high-school students**
Sharif University of Technology
Taught an intuitive introduction to differential geometry.

Teaching (continued)

2012–2014 **Mathematical olympiad preparation**
Taught geometry and special topics in mathematics to high-school students.

Academic service

2014 **Invited grader, First Iranian Geometry Olympiad**

Honours and funding

2023–2027 **PhD scholarship**
University of Sussex

2021–2022 **EUR M&CS scholarship**
Université Sorbonne Paris Nord

2016 **Full participation scholarship**
Workshop on Symplectic Field Theory VIII, Humboldt-Universität zu Berlin

2015 **Full participation scholarship**
School and Workshop on Complex Analysis, Geometry and Dynamics, ICTP

2015 **Full participation scholarship**
CIMPA–NIMS Research School on Dynamical Systems, Daejeon

2011 **Bronze medal, National Mathematical Olympiad**
Iran

Conferences and schools attended

9 Oct 2026 **Fourth London Probability Day**
University College London, UK

15–19 Jun 2026 **Integrable systems: regularity, non-commutativity and random matrix theory**
ICMS, Bayes Centre, Edinburgh, UK

Jan 2026 **London Probability Day and Tangents and New Normals**
Queen Mary University of London, UK

Dec 2024 **XX Brunel–Bielefeld Workshop on Random Matrix Theory and Applications**
Brunel University London, UK

Oct 2024 **Algebraic aspects of random matrices**
CIRM, Marseille, France

10–18 Jun 2024 **Random matrix and free probability summer school**
Toulouse, France

22–26 Aug 2020 **Fourth IPM Advanced School on Computing: Artificial Intelligence**
Online; IPM, Tehran, Iran

Winter 2018 **Topology and Dynamics in the Swiss Alps**
Switzerland

8–12 Aug 2016 **Workshop on Symplectic Field Theory VIII**
Humboldt-Universität zu Berlin, Germany

21 Sep–3 Oct 2015 **School and Workshop on Complex Analysis, Geometry and Dynamics**
ICTP, Trieste, Italy

10–14 Aug 2015 **Invariants in Low Dimensional Geometry**
Gazi University, Ankara, Turkey

Summer 2015 **CIMPA–NIMS Research School on Dynamical Systems: Examples, Billiards, and the 3-Body Problem**
Daejeon, South Korea

2014–2015 **Third Frontiers in Mathematical Sciences conference**
IPM, Iran

28 Jun–25 Jul 2014 **Summer School on Analysis and PDE**
IPM, Iran

2013–2014 **Second Frontiers in Mathematical Sciences conference**
Sharif University of Technology, Iran